

# STA623 - Bayesian Data Analysis

## Oral examination on your assignment: information for students

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|        |                                                                                              |
|--------|----------------------------------------------------------------------------------------------|
| Weight | 25% of the overall course grade (separate from the 25% for the written assignment)           |
| When   | December 2026; ~1–2 weeks after the assignment due date. Your time slot will be sent to you. |
| Length | 10-15 minutes                                                                                |
| Where  | online, using MS Teams                                                                       |

### What it is

A 10-15 minute conversation with the course instructor / examiner about **the assignment report you submitted**. The aim is to find out how well you understand your own analysis: why you made the choices you made, what your code does, what your results mean, and how you would reason if things were different.

It is not a test of new material, and there are no trick questions. If you did the analysis yourself and understood it, you are well prepared already.

### What you will have

- A copy of **the PDF report and the .qmd file you submitted** - keep a copy of your submission. You may look at them at any time.
- Pen and paper, for writing or sketching while you think.
- No other notes, and no laptop or phone.

### What will happen (this is a guide only)

| Part    | Time  | What you will be asked                                                                      |
|---------|-------|---------------------------------------------------------------------------------------------|
| Summary | 2 min | Summarise your findings in two minutes, as you would for a district health management team. |

| Part                      | Time    | What you will be asked                                                                              |
|---------------------------|---------|-----------------------------------------------------------------------------------------------------|
| Your data and decisions   | 3 min   | Questions about one part of your data checking, variable coding or priors.                          |
| Your code and computation | 4 min   | Questions about your own NIMBLE code (shown on screen) and the computational problem you described. |
| Your checks and results   | 3 min   | Questions about your model checks, your model comparison or your answers to the research questions. |
| What if                   | 1–2 min | One question about how your analysis or conclusions would change in a different situation.          |

The examiner will point to specific tables, figures and lines of code in your report. Every student gets questions about their own work, so your questions will differ from your classmates’.

## How it is marked

Four criteria, each scored from 0 to 5:

1. **Ownership of the work:** you can explain your own code and where the numbers in your report come from.
2. **Data and modelling decisions:** you can give reasons for your choices, the assumptions behind them, and the alternatives.
3. **Computation and checking:** you understand your MCMC diagnostics, the computational problem you met, your posterior predictive checks and your LOO comparison.
4. **Interpretation and judgement:** you can explain your results in plain terms, with their uncertainty, and reason about what would change in a different situation.

The oral is marked on **what you show in the conversation**, not on the quality of your written report, which has already been marked. If you realise in the oral that something in your report was wrong, **say so, and explain what you would do instead**: recognising and explaining a mistake counts in your favour.

Answers that stay general (“the model converged well”) score less than answers that refer to your own data, code and numbers (“the R-hat for  $\sigma$  was 1.08 in my first run, because...”).

## The oral and the use of AI tools

The assignment allowed AI tools, if declared. The oral does not change this. You will be asked to explain your work, including code an AI tool helped you write, and you may be asked about the example in your AI-use statement. Anything in your report that you cannot explain will count against you in the oral.

If there are serious doubts that the submitted report is your own work, this will trigger UNIMA's academic integrity process.

## How to prepare

- **Re-read your report and your .qmd file.** Make sure you can explain every model, every major choice and every key number, and where in the code each comes from.
- **For each decision in your decision log, be ready to say why**, and what the alternative would have been.
- **Practise explaining your results in plain language**, for example the hypoxaemia effect in deaths per 100 children.
- **Think about “what if” questions**: what would change with more hospitals, different priors, or a different kind of missing data?

During the oral, take your time. It is fine to pause, to look in your report, to write something down, or to ask the examiner to repeat or rephrase a question. Clear, simple English is all that is needed.

## Practicalities

- The oral will be **audio-recorded** for assessment purposes. The recording will be deleted as soon as the course grades have been finalised by UNIMA.
- If you need **reasonable adjustments** (for example extra time), let the course instructor know - at least a week in advance.
- If you cannot attend your slot for a valid reason, contact the course instructor as early as possible to arrange another one.